Road damage is easy to miss.

Cracks. Potholes.
Details that need a closer look.



Frankie Fouganthin / Pothole in an asphalt pavement / cropped
commons.wikimedia.org/wiki/File:Pothole_in_an_asphalt_pavement.jpg
Photo adaptation: creativecommons.org/licenses/by-sa/4.0/

Meet RoadScan.

AI suggests possible damage. You check the result.

Upload → Detect → Review

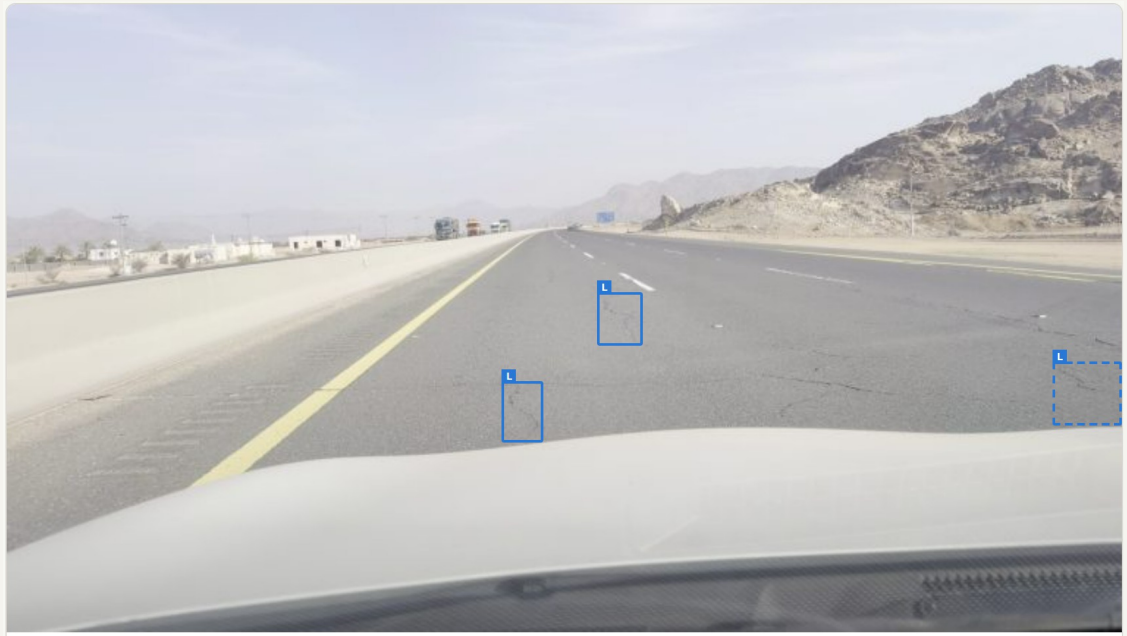
Start with a road image.

Get damage boxes and labels.

Correct the result and save your review.

Model suggestions on real road images

YOLO26s Main / saved project predictions



Original

found 2 of 6 · 1 false



Original

found 1 of 5 · 1 false

Dashed boxes retain the source’s false-alarm flags.

Selected project examples, not an independent test.

Models we explored

19 recorded detector choices, including training variants

YOLO detectors

YOLO11n / YOLO11s / YOLO11m
YOLOv8m
YOLO26n / YOLO26s

Transformer detectors

RT-DETR L / RT-DETR X

Classifier baseline

ResNet18 · image-level labels

Main: YOLO26s 1024 → Road v1 / A / v2b

Built around our YOLO26s models.

Recorded RDD split: ~27k training · 5,758 validation · 5,758 test

01

Prepare data

RDD road images
Audit splits and labels

02

Transfer learning

Pretrained YOLO26
Adapt to road damage

03

Run experiments

n / s · 640 / 1024
Augmentation + labels

04

Choose Main

1024-pixel input
Best validation: epoch 84

Later road fine-tunes: Road v1 / A / v2b

What Main achieved

YOLO26s Main · 1024 px · Ultralytics RDD test (5,758 images)

63.04%

mAP50

33.01%

mAP50–95

What the 640-pixel RDD experiments showed

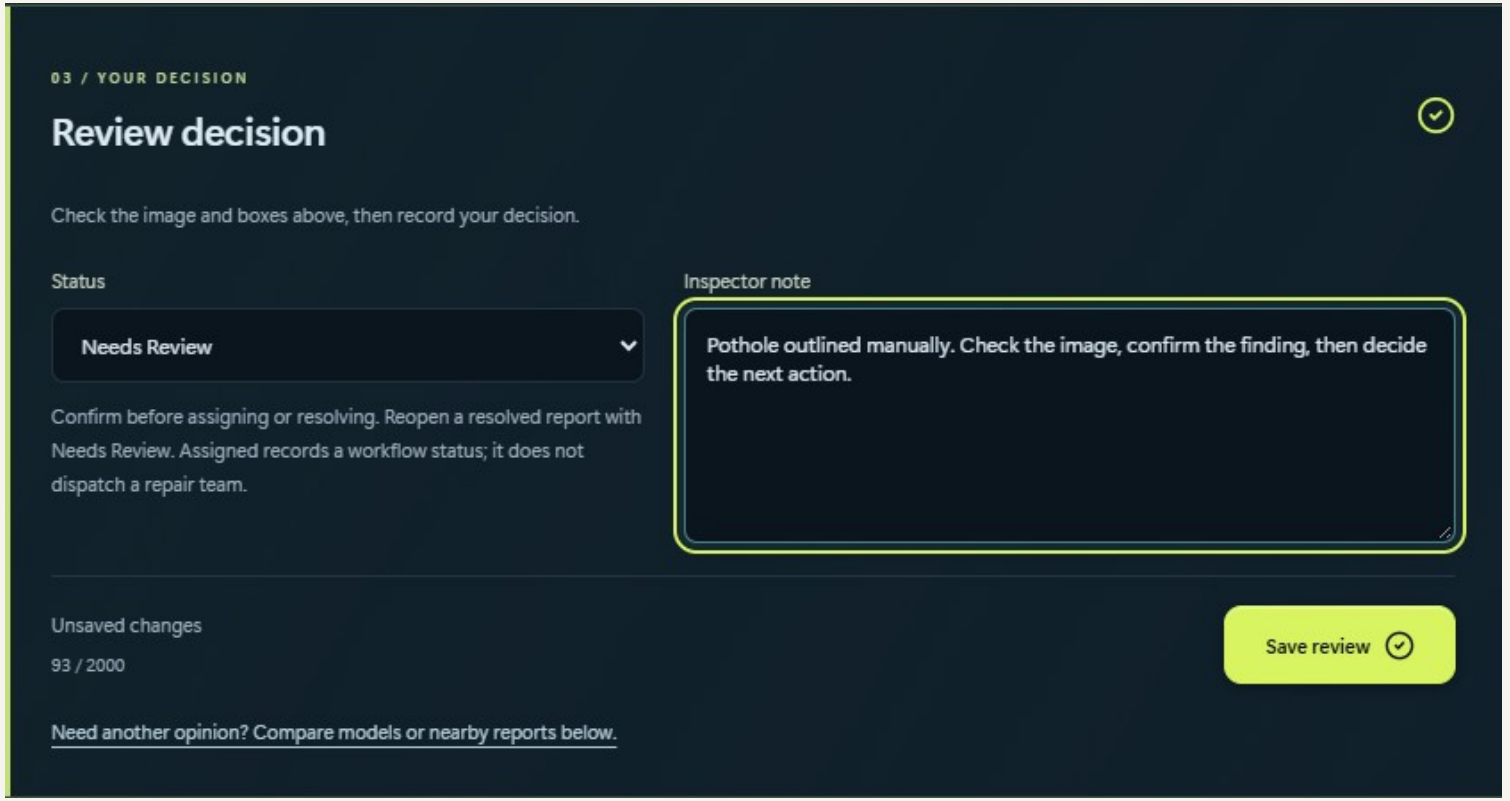
Nano → Small
55.99% → 59.43% mAP50

No augmentation → Standard
46.93% → 59.43% mAP50

Four damage classes · historical results · near-duplicate leakage reported

Review it before you trust it.

Check the boxes.
Correct mistakes.
Save the decision.



Actual review controls

How RoadScan works

Detect	→ Prioritize	→ Find repeats	→ Review queue
YOLO boxes and confidence	Type, area, count and confidence	DINOv2 crops FAISS + GPS radius	Ranked reports Human decision

Priority = 100 × (0.40T + 0.25A + 0.20N + 0.15C)

Duplicate branch: optional FAISS · threshold uncalibrated · end-to-end acceptance pending

Live demo · 2.5 minutes · the panel chooses the input

See the damage.
Keep the decision.

Damage detection with a reviewable result.
A local test set is still missing.
Next: evaluate on unseen local roads.

Feras Alyamani / Ahmed Salamah / Ahmed Alhejaili

Road-model results, with context

Road variants / 343-frame selection set

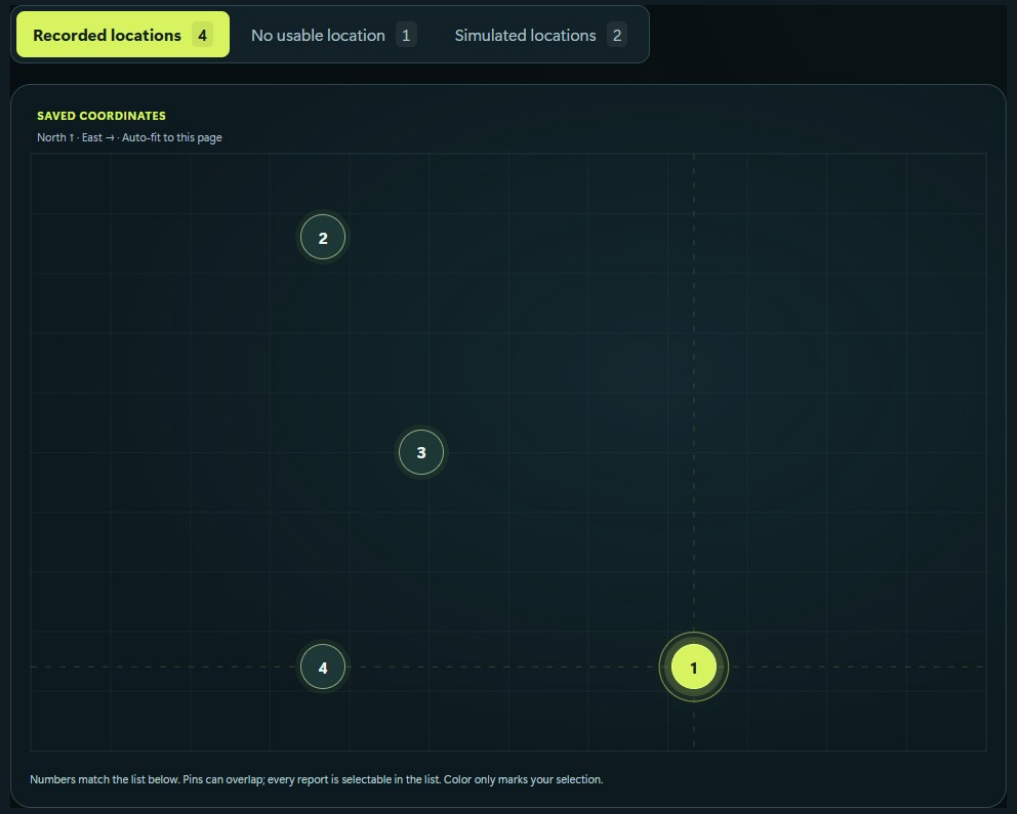
YOLO26s variant	mAP50	mAP50–95
Road v1	46.74%	31.26%
Road A	46.80%	28.98%
Road v2b	48.61%	29.37%

Selection scores, not an independent accuracy test.

Main / separate RDD test: 63.04% mAP50, 33.01% mAP50–95

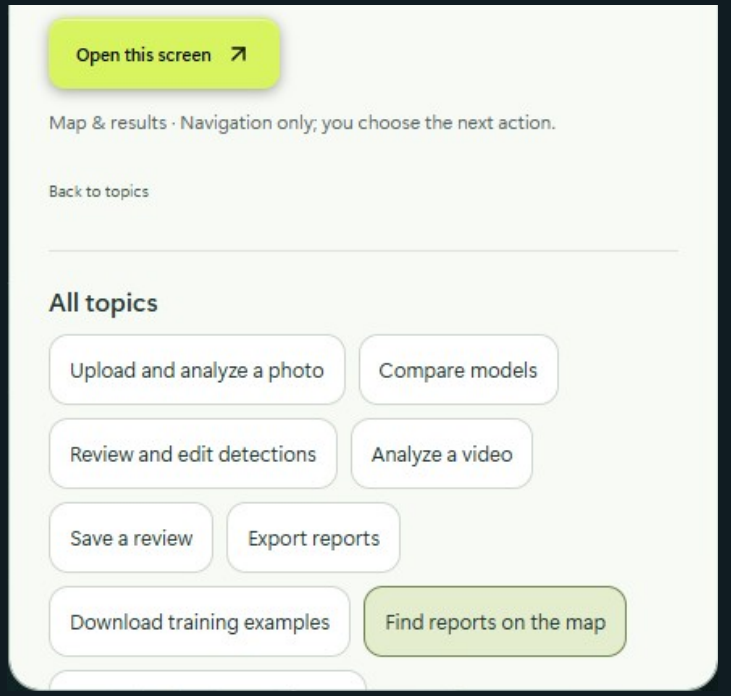
Find the evidence when you need it.

A map linked to your reports



Recorded, missing and simulated locations stay distinct.

Help in English and Arabic



Clear steps. You choose the action.

Experiments that shaped the system

Historical results across separate experiments

Experiment	Reported evidence (mAP: %)	Decision / limit
E1 · Classifier	Any-damage F1: ResNet18 .893 / Main .929	Detector adds localization
E2 · Size	640 px: n 55.99 / s 59.43 mAP50	Small; speed trade-off
E3 · Augmentation	No aug. / standard: 46.93 / 59.43 mAP50	Keep standard augmentation
E4 · Country shift	India excluded / included: 8.86 / 39.25 mAP50	Local holdout still missing
E5 · Embeddings	DINOv2 + GPS: .980 F0.5	Simulated; app uncalibrated

Different protocols; no common ranking. Road’s 343 frames are selection data.